UNIT – II

Strengthening Mechanisms

Subject : Mechanical Behaviour of Materials (23A03504c)

> Syllabus: Cold working, grain-size, solid-solution, martensitic, precipitation, dispersion and fibre strengthening with examples ; yield-point phenomenon, strain ageing and dynamic strain ageing.

1. Principle of Strengthening

Plastic deformation occurs by dislocation motion – so strengthening = restricting dislocation motion. Every mechanism introduces obstacles that raise the stress needed to move dislocations ($\tau = \tau_0 + \Delta\tau$), at the expense of ductility and toughness (usually).

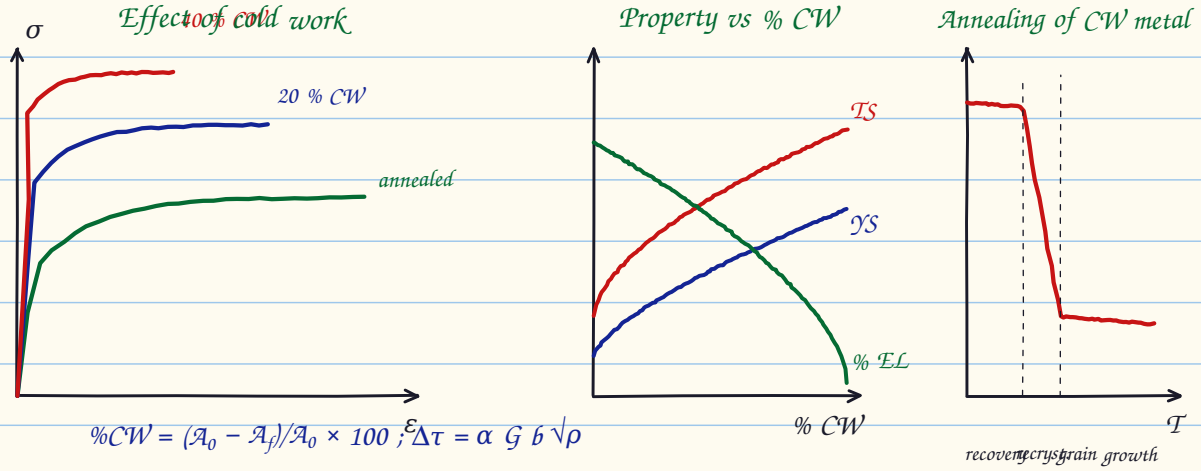
Obstacle	Mechanism	Technique
Other dislocations	forest hardening	cold work
Grain boundaries	pile-up blocked	grain refinement
Solute atoms	lattice friction / atmospheres	alloying
Fine precipitates	shear or bypass	age hardening
Hard dispersoids	Orowan bypass	powder metallurgy (ODS)
Fibres / reinforcement	load transfer	composites
Phase / BCT lattice	supersaturation, high ρ	quench (martensite)

- Strengthening effects are roughly additive : $\sigma_y = \sigma_0 + \Delta\sigma_{ss} + \Delta\sigma_{gb} + \Delta\sigma_\rho + \Delta\sigma_{ppt}$ (sometimes by root-sum-square).
- Requirement : a good strengthening process should leave ductility, toughness and high-temperature stability acceptable.

2. Cold Working (Strain Hardening / Work Hardening)

Plastic deformation below the recrystallisation temperature ($\approx 0.4\ T_m$) – rolling, drawing, forging, bending.

$$\%\ CW = (A_0 - A_f)/A_0 \times 100$$



- ***Mechanism*** *: plastic strain multiplies dislocations (Frank–Read sources) – ρ rises from 10^6–10^8 to 10^{11}–10^{12} cm^{-2} ; dislocations interact, tangle, form pile-ups, cells and locks (Lomer–Cottrell), so further slip needs a higher stress.*

$$\Delta\tau = \alpha\ G\ b\ \sqrt{\rho}\ \text{(Taylor, } \alpha \approx 0.2\text{–}0.5) \quad \sigma_T = K\varepsilon_T^{\,n}$$

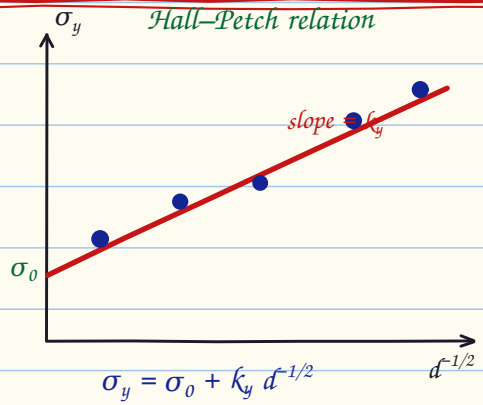
- *Effects of cold work : YS ↑↑, TS ↑, hardness ↑, ductility ↓, electrical conductivity slightly ↓, corrosion (stress) susceptibility ↑,* ***anisotropy*** *(texture, banding), stored energy ≈ 5 – 10 % of the work retained.*
- *Strain hardening exponent n : FCC (Cu, brass, stainless) n = 0.3–0.5 ; low-C steel 0.2 ; HCP 0.1–0.2. Stage hardening of single crystals : I easy glide, II linear (θ ≈ G/300), III parabolic (dynamic recovery).*
- ***Limit*** *: hardening cannot be increased indefinitely (ductility exhausted) ; it is lost on annealing (recovery → recrystallisation → grain growth).* ***Recrystallisation temperature*** *≈ 0.3–0.5 T_m ; decreases with % CW, purity, small prior grain size. Typical : pure Cu 200 °C, Fe 450 °C, Al 80 °C, W 1200 °C.*

Ex 1. Wire drawn from 5 mm to 4 mm dia : A_0 = 19.63, A_f = 12.57 mm^2 ⇒ % CW = 36 %. For ρ rising to 10^{15} m^{-2} in Cu, $\Delta\tau = 0.5 \times 48\times10^{9} \times 2.55\times10^{-10} \times 3.16\times10^{7}$ = 193 MPa.

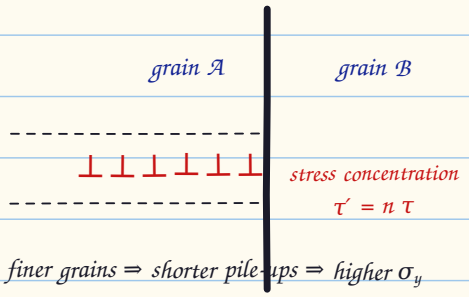
- *Applications : cold-drawn wires, cold-rolled sheet (strain-hardened tempers H12–H18 Al), copper conductors, cold-worked stainless steel springs, shot peening (surface).*

3. Grain-Size Strengthening (Hall–Petch)

Hall–Petch relation



Pile-up at a grain boundary



Grain boundaries stop slip (adjacent grain has a different orientation) – dislocations pile up against them. The stress concentration of n piled-up dislocations ($\tau' = n\tau$) must reach the stress needed to activate a source in the next grain :

$$\sigma_y = \sigma_0 + k_y\, d^{-1/2} \quad \text{(Hall 1951, Petch 1953)}$$

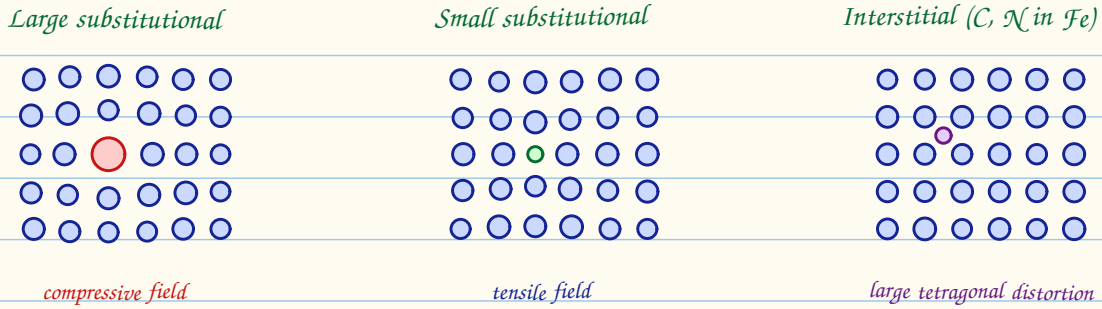
- σ_0 = friction stress (lattice resistance to dislocation motion) ; k_y = locking parameter (measure of how effectively grain boundaries block slip) ; d = average grain diameter. Valid for d from ≈ 1 mm to ≈ 20 nm.
- k_y ($MPa \cdot m^{1/2}$) : low-C steel ≈ 0.6 ; Cu 0.11 ; Al 0.07 ; Ti 0.4 ; Mg 0.28 ; Ni 0.16. Increase of k_y for BCC (strong locking by atmospheres).
- Fine grain also improves toughness and lowers DBTT (unlike other methods) – so the best way to raise strength and toughness together (HSLA steels, controlled rolling). Grain refinement methods : rapid solidification, inoculation (Ti-B in Al), recrystallisation after cold work, thermo-mechanical controlled processing, microalloying (Nb, V, Ti), severe plastic deformation (ECAP).
- ***Inverse Hall–Petch*** : for d < 10–20 nm (nano-crystals) strength falls – grain-boundary sliding dominates. ***Limits*** : at high temperature (> 0.5 T_m) fine grains are weaker (grain-boundary sliding, creep – see Unit V).

Ex 2. Low-carbon steel σ_0 = 100 MPa, k_y = 0.6 $MPa \cdot m^{1/2}$: d = 25 µm ⇒ $d^{-1/2}$ = 200 $m^{-1/2}$ ⇒ σ_y = 100 + 0.6×200 = 220 MPa ; d = 4 µm ⇒ 500 ⇒ 400 MPa ; d = 1 µm ⇒ 700 MPa.

- ASTM grain size : $N = 2^{n-1}$ grains per square inch at 100× ; larger n ⇒ finer grains.

4. Solid-Solution Strengthening

Solute atoms in the matrix (substitutional or interstitial) make the alloy stronger than the pure metal because they distort the lattice and interact with dislocations.



- **Sources of interaction** : (1) size misfit $\varepsilon = (1/a)(da/dc)$ – strain field interacts with the hydrostatic part of a dislocation's field (edge) ; (2) modulus mismatch ; (3) electrical / chemical (Suzuki segregation to stacking faults) ; (4) order effects. Large solutes sit in the tensile region below an edge dislocation, small ones in the compressive region above.
- **Cottrell atmosphere** : solute accumulates around dislocation cores (C, N in Fe ; interstitials) and lowers their energy by ≈ 0.5 eV — the dislocation is locked ⇒ higher stress to break away (upper yield point, strain ageing).

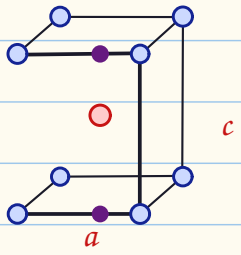
$\Delta\tau \propto G\,\varepsilon^{3/2}\,c^{1/2}$ (Labusch/Fleischer, dilute) ; or $\Delta\tau \propto c^{2/3}$; $\Delta\tau = G\,\varepsilon\,c$ (Mott–Nabarro)

- Strengthening increases with : solute concentration c ; misfit ε (Hume-Rothery rule : size difference < 15 % for extensive solubility) ; interstitials are 10–100 times as effective as substitutional (non-spherical distortion, tetragonal field interacts with shear stress of screw also).
- Examples : brass (Cu–Zn), bronze (Cu–Sn), Cu–Ni (monel), Al–Mg (5xxx alloys), stainless steel (Cr, Ni, N), Mn/Si in ferrite. Approx. rise of ferrite yield strength per wt % : Si ≈ 80 MPa ; Mn ≈ 30 MPa ; Cu ≈ 40 MPa ; P ≈ 700 MPa ; interstitial C+N ≈ 4000 MPa/wt % (limited by solubility).
- Effect on properties : YS, TS ↑, ductility ↓, conductivity ↓ strongly, annealed alloys work-harden faster. Maximum strengthening near 50 % composition for complete-solubility alloys (Cu–Ni). Temperature : solute strengthening persists to high T (≈ 0.6 T_m) – basis of solid-solution superalloys (Inconel 625).

Ex 3. Cu has YS ≈ 70 MPa ; 70-30 brass (30 % Zn) annealed has YS ≈ 100 MPa, TS 325 MPa ; Cu + 10 % Ni annealed TS ≈ 270 MPa and Cu + 30 % Ni ≈ 380 MPa.

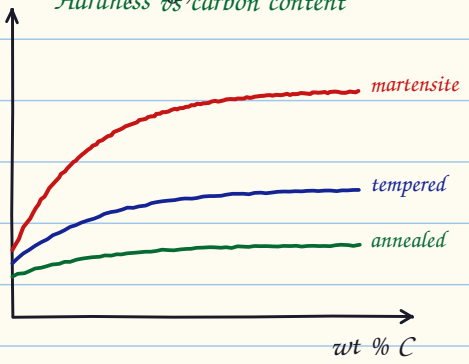
5. Martensitic Strengthening

Body-centred tetragonal (martensite)



$c/a = 1 + 0.045$ (wt % C) ; C at octahedral sites

Hardness vs carbon content



Martensite forms when austenite (FCC γ-Fe containing carbon) is cooled so rapidly (quenched) that carbon cannot diffuse out as cementite ; the lattice shears (diffusionless, athermal, ≈ speed of sound) into a body-centred tetragonal (BCT) structure supersaturated with carbon.

- **Sources of strength** : (1) interstitial carbon – very potent solid-solution hardening ; (2) very high dislocation density (10^{11}–10^{12} cm^{-2}) and twins formed by the shear transformation ; (3) extremely fine lath/plate substructure (boundaries – Hall–Petch) ; (4) carbon segregation to dislocations and later fine carbide precipitates (in tempering).

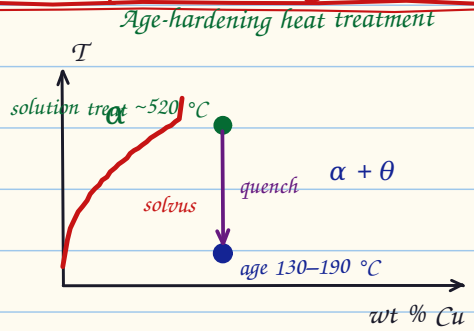
$$c/a = 1 + 0.045 \text{ (wt \% C)} \quad M_s(°C) = 539 - 423\,C - 30.4\,Mn - 17.7\,Ni - 12.1\,Cr - 7.5\,Mo$$

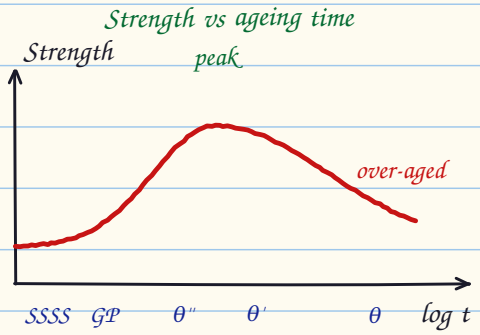
- Morphology : lath (low C < 0.6 %, high dislocation density, tougher) ; plate (high C > 1 %, twinned, brittle, many micro-cracks) ; mixed between. Retained austenite for M_f < room T (high C).
- Hardness of martensite depends mainly on carbon : HRC ≈ 35 (0.2 % C), 55 (0.4 %), 65 (≥ 0.6 %). Yield strength may be 1500–2000 MPa but brittle ⇒ tempering (150–650 °C) – carbon precipitates as carbides (ε-carbide → Fe_3C), lath recovers : strength ↓, toughness ↑ (tempered martensite = best combination).
- Hardenability (depth) governed by alloy content (Mn, Cr, Mo, Ni) – Jominy test ; quenching cracks and distortion are limitations. Related : bainite (carbide +

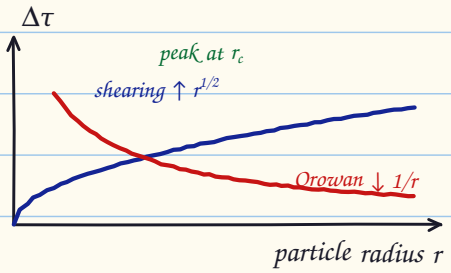
ferrite, ≈ 0.5 hardness), **TRIP** and dual-phase steels, maraging steels (Fe–18Ni martensite aged with Ni_3Ti, 2000 MPa).

Ex 4. 0.4 % C, 0.8 % Mn steel : $M_s = 539 - 169.2 - 24.3 = 345\ °C$; $c/a = 1 + 0.045 \times 0.4 = 1.018$.

6. Precipitation (Age) Hardening







SSSS → GP zones → θ″ → θ′ → θ (Al_2Cu)

coherent → semi-coherent → incoherent

Alloys whose solubility of the solute decreases with temperature can be strengthened by a controlled precipitate dispersion (Wilm 1906 – duralumin). Examples : Al–Cu (2xxx), Al–Mg–Si (6xxx), Al–Zn–Mg (7xxx), Cu–Be, Ni-base superalloys, maraging steel, PH stainless steels (17-4).

Heat treatment steps

- (1) **Solution treatment** : heat into single-phase α region (≈ 520 °C for Al–4.5Cu) and soak to dissolve the second phase. (2) **Quench** rapidly – supersaturated solid solution (SSSS) is retained. (3) **Ageing** : natural (room T) or artificial (130–190 °C) – decomposition proceeds.
- **Precipitation sequence** (Al–Cu) : SSSS → GP zones (coherent Cu-rich discs, 1–2 atom layers) → θ″ (coherent, tetragonal) → θ′ (semi-coherent) → θ (Al_2Cu) (incoherent equilibrium). Strength maximum at θ″ + θ′ ; **over-ageing** (coarse θ) lowers strength.

Strengthening mechanisms

- **Coherency strain hardening** (small zones) : lattice misfit strains impede dislocations.

- **Particle shearing** (small, coherent, $r < r_c$) : dislocations cut through – extra energy for new interface area (surface hardening), order hardening (γ' in Ni alloys), modulus misfit, stacking-fault differences. $\Delta\tau \propto \sqrt{(f\,r)}$ – increases with r.
- **Orowan bypassing** (large, incoherent, $r > r_c$) : dislocation bows between particles and leaves loops :

$$\Delta\tau = G\,b / \lambda \quad \lambda \approx 2r(\sqrt{(\pi/4f)} - 1) \text{ (decreases as particles coarsen)}$$

- Peak strength at the critical radius r_c where the shearing and Orowan stresses are equal.

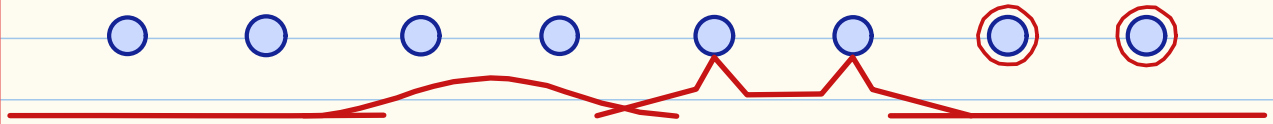
Ex 5. Al alloy : $G = 26$ GPa, $b = 2.86$ Å ($Gb = 7.4$ N/m), $f = 0.02$, $r = 10$ nm : $\lambda = 2(10)(\sqrt{(\pi/0.08)} - 1) = 20 \times 5.27 = 105$ nm $\Rightarrow \Delta\tau = 7.4/105\times10^{-9} =$ 70 MPa $\Rightarrow \Delta\sigma \approx M\,\Delta\tau = 3.06 \times 70 \approx$ 215 MPa. If particles coarsen so $\lambda = 1$ µm : $\Delta\tau =$ 7.4 MPa (over-aged).

- Requirements for an age-hardenable system : appreciable maximum solid solubility, solubility ↓ with T, an alloy that forms coherent transition precipitates. Natural ageing (T4) vs artificial (T6) ; duralumin hardens at RT in days. Dis-advantages : loses strength above ≈ 150–200 °C (coarsening, dissolution) ; welding zones soften (HAZ).
- **Related** : Spinodal decomposition (Cu–Ni–Fe, alnico) ; strain-age (double) treatments ; precipitation hardened Ni-superalloys retain strength to 0.8 T_m via ordered γ' $Ni_3(Al,Ti)$.

7. Dispersion Strengthening

Orowan bowing around strong particles : $\tau = G\,b / \lambda$

loops accumulate ⇒ rapid work hardening



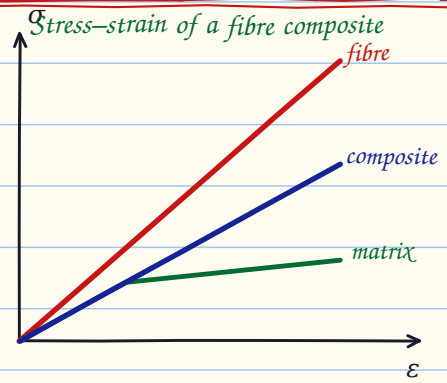
(a) approach (b) bows between (c) pinches off (d) leaves loops

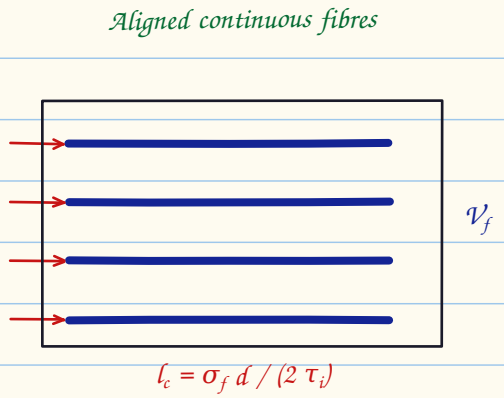
Very fine, stable, insoluble second-phase particles (oxides, carbides, nitrides – Al_2O_3, ThO_2, Y_2O_3) introduced by powder metallurgy / mechanical alloying / internal oxidation / in-situ reaction. Particle size 10–100 nm, volume fraction 1–15 %,

spacing $\lambda \approx 0.1$–$0.5\ \mu m$.

- Strengthening by **Orowan bowing** : $\tau = G\, b / \lambda$; work hardening rate is high (loops left behind).
- **Key difference from age hardening** : particles do not dissolve or coarsen at high temperature (no solubility) ⇒ strength retained up to $\approx 0.9\ T_m$, excellent creep resistance. Strength at room temperature is lower than age-hardened alloys, and ductility/toughness lower.
- Materials : SAP (sintered Al powder with Al_2O_3), TD-nickel (Ni–2 % ThO_2), ODS alloys (MA754, MA6000, PM2000 Fe-Cr-Al–Y_2O_3), dispersion-strengthened copper (Cu–Al_2O_3 "Glidcop" – contact tips, resistance welding electrodes), W–ThO_2 filaments, cemented carbides (WC-Co).
- Conditions for effectiveness : high G of particles, uniform distribution, small size at fixed f ($\lambda \propto r/\sqrt{f}$), particle–matrix interface strong and no reaction ; ratio of Orowan stress to Peierls stress.
- Example : Cu with 0.5 vol % Al_2O_3 ($\lambda \approx 0.1\ \mu m$) has 380 MPa YS with conductivity 90 % IACS and retains it at 900 °C, while age-hardened Cu–Be softens above 300 °C.

8. Fibre Strengthening (Composites)





Strong, stiff fibres embedded in a ductile matrix : the matrix transfers load to the fibres through the interface shear stress, fibres carry most of the load. Fibres : glass, carbon, boron, Kevlar (aramid), SiC, Al_2O_3, steel wire ; matrices : polymers (epoxy), metals (Al, Ti), ceramics.

- **Continuous aligned fibres, load parallel** (isostrain, rule of mixtures) :

$E_c = E_f V_f + E_m V_m$ $\sigma_c = \sigma_f V_f + \sigma_m' V_m$ (σ_m' = matrix stress at fibre-fracture strain)

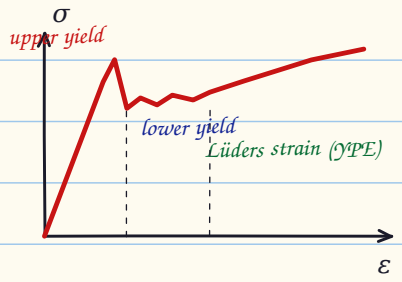
- **Transverse load** (isostress) : $1/E_c = V_f/E_f + V_m/E_m$ – much lower ; strongly anisotropic ⇒ use cross-ply laminates.
- **Critical fibre length** (fibre fully loaded to its strength) : $l_c = \sigma_f d / (2 \tau_i)$. Fibres with $l < l_c$ do not reach fracture stress ; discontinuous (short) fibres : $\sigma_c = \sigma_f V_f(1 - l_c/2l) + \sigma_m' V_m$ for $l > l_c$; random orientation gives ≈ 0.2–0.4 of the aligned value.
- **Matrix role** : bonding, load transfer, protects fibres, separates them (crack stopper) ; ductile matrix with brittle fibres gives toughness through fibre pull-out and debonding.

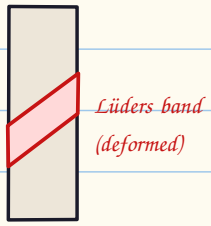
Ex 6. Glass/epoxy, $V_f = 0.5$: $E_f = 72$, $E_m = 3$ GPa ⇒ E_c (parallel) = 36 + 1.5 = 37.5 GPa ; transverse : $1/E = 0.5/72 + 0.5/3 = 0.174$ ⇒ 5.8 GPa. $\sigma_f = 2000$ MPa, $\sigma_m' = 60$ MPa : $\sigma_c = 1000 + 30 =$ 1030 MPa. $l_c = (2000 \times 10\ \mu m)/(2 \times 20\ MPa) =$ 0.5 mm.

- Other composite types : particulate (cermets, concrete), laminar (plywood, clad metals), MMC (SiC/Al – engine cylinders), CMC (SiC/SiC – turbine hot parts), carbon/carbon (brakes). Specific strength (σ/ρ) and specific stiffness (E/ρ) are very high – aerospace, sports goods, wind blades.
- **Similar principle in steel** : ferrite-pearlite, cementite lamellae strengthening ; reinforced concrete ; whiskers (SiC).

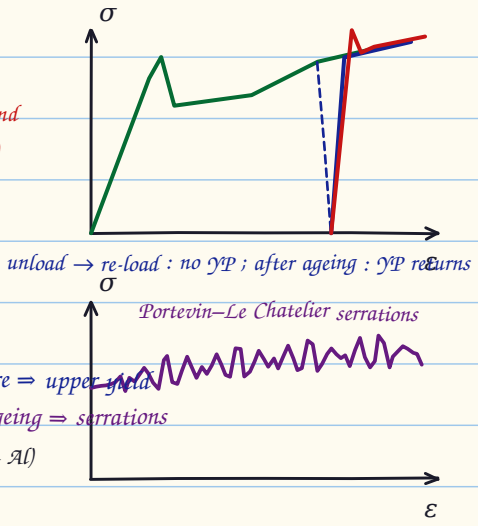
9. Yield Point Phenomenon

Yield-point phenomenon





Strain ageing



Low-carbon steels (and Mo, Ti, Mo alloys with interstitials) show a sharp upper yield point followed by a drop to a lower yield point and plastic flow at nearly constant stress (**yield point elongation**, 1–3 % – Lüders strain) before strain hardening begins.

- **Lüders bands** : localised bands of plastic deformation (≈ 45° to axis) nucleate at stress concentrations and propagate along the gauge length at constant load ; the whole specimen yields when the band front has traversed it. Surface appearance : stretcher-strain marks / "worms" in deep-drawn sheet – unacceptable in automobile panels.
- **Explanation (Cottrell–Bilby)** : C and N atoms segregate to dislocation cores forming atmospheres that anchor them ; a high stress is needed to tear the dislocations away (upper yield) ; once free they move at a lower stress (lower yield). **Johnston–Gilman** : initial mobile dislocation density is very low, so to achieve the imposed strain rate each must move fast ; $v \propto \tau^m$ ($m \approx 20$–30) ⇒ the stress needed is high at first and falls as dislocations multiply ($\dot{\varepsilon} = \rho_m\, b\, v$).
- **Removal** : (i) temper (skin-pass) rolling 0.5–2 % – generates mobile dislocations ; (ii) Al-killed or Ti-stabilised steel (N tied up), IF steel (interstitial-free) ; (iii) forming soon after temper rolling ; (iv) levelling by roller.
- Factors : more pronounced for fine grain, low temperature, high strain rate ; absent in pure FCC metals. Observed in Fe, Mo, Ti, Cd, Zn, Al–Mg, and in single crystals of Ge, Si, LiF.

10. Strain Ageing & Dynamic Strain Ageing

Static strain ageing

- Sequence : (1) strain steel beyond the yield point to A ; (2) unload and **immediately reload** – smooth elastic–plastic curve, no yield point (dislocations free) ; (3) **age** – hold at room temperature for days or at 100–200 °C for minutes – then reload : yield point returns at a higher stress (σ_y ↑, ductility ↓).
- Cause : interstitial C and N diffuse back to the dislocations (pinning, Cottrell) ; Harper's law : fraction pinned $f = 1 - \exp[-(Kt)^{2/3}] \propto (D\,t)^{2/3}$; strain ageing activation energy ≈ that of C diffusion in ferrite (80 kJ/mol).
- **Consequences** : stretcher strains reappear in stored sheet (use within weeks), hardness ↑ and toughness ↓ (↑ DBTT) of cold-worked structural steel after ageing ; **blue brittleness** at 200–300 °C (blue oxide colour) – ductility minimum.

Dynamic strain ageing (DSA)

- Strain ageing that occurs during deformation, when solute diffusion rate is comparable to the speed at which dislocations are arrested at obstacles (waiting time t_w ≈ diffusion time) : occurs in a window of temperature and strain rate (e.g. 100–300 °C for C/N in steel, room temperature for Mg in Al, 400–600 °C for many alloys).
- **Portevin–Le Chatelier (PLC) effect** : repeated yield drops – **serrated stress–strain curve** (types A – periodic locking, B – oscillating, C – sudden drops) due to repeated pinning and unpinning of dislocations ; visible bands in Al–Mg alloys (5083, 5182 – surface roughness "Lüders-like" defects).
- Characteristic signs : negative strain-rate sensitivity ($\partial\sigma/\partial \ln \dot{\varepsilon} < 0$), increased work-hardening rate and flow stress, ductility minimum, strength peak with temperature (bump in σ–T curve), and serrations above a critical strain ε_c that rises with strain rate and falls with temperature.
- Avoid : by changing composition (less Mg/N), heat treatment (precipitation of solutes), avoiding DSA temperature window in forming, texture. Practical : structural steel at ≈ 250 °C (reactor pressure vessel) loses ductility.

11. Examples & Comparison of Mechanisms

Mechanism	Example	Principal obstacle	Limit
Cold work	Cu wire, Al H14 sheet	forest dislocations	annealed away at 0.4 T_m
Grain size	HSLA steel, ultrafine Al	grain boundaries	loses at high T (creep)
Solid solution	brass, stainless, Al–Mg	solute atoms	moderate gain ; < 0.6 T_m
Martensite	quench & tempered 4340	carbon, ρ, laths	brittle ; tempers away ≈ 200 °C
Precipitation	2024-T6, 7075, Cu–Be, Inconel 718	coherent/shearable particles	coarsen/dissolve at ≈ 0.5 T_m
Dispersion	TD-Ni, ODS, SAP, Glidcop	inert oxides (Orowan)	lower RT strength
Fibre	GFRP, CFRP, SiC/Al	fibre load-carrying	anisotropy, cost

- **Combination in practice** : HSLA steel ($\sigma_y \approx 450$ MPa) : $\sigma_0 \approx 50$ + solid solution (Mn, Si) 90 + grain size ($d = 5\ \mu m$) 160 + dislocations 60 + Nb/V carbide precipitates 90 ⇒ ≈ 450 MPa. Ni superalloy : solid solution + γ′ (ordered) + carbides on boundaries + (single crystal, no boundaries).

12. Additional Solved Problems

P1. A Cu rod 12.7 mm dia is cold drawn to 10.2 mm : $\%CW = (126.7 - 81.7)/126.7 = 35.5\ \%$.

P2. Brass : $\sigma_y = 70$ MPa at $d = 0.016$ mm and 140 MPa at $d = 0.0008$ mm ⇒ $d^{-1/2} = 7.9$ and 35.4 $mm^{-1/2}$ ⇒ $k_y = 70/27.5 = 2.55\ MPa{\cdot}mm^{1/2}$; $\sigma_0 = 70 - 2.55 \times 7.9 = 49.9$ MPa.

P3. Martensite M_s for 0.8 % C steel : $539 - 423(0.8) = 200.6$ °C ($M_f \approx -50$ °C ⇒ retained austenite at RT).

P4. Orowan stress for Ni alloy $\lambda = 50$ nm, $G = 76$ GPa, $b = 2.49$ Å : $\Delta\tau = 76\times10^{9}\times2.49\times10^{-10}/(50\times10^{-9}) = 378$ MPa.

P5. Fibre composite : Al–SiC, $V_f = 0.3$, $E_f = 400$, $E_m = 70$ GPa : $E_c = 120 + 49 = 169$ GPa.

Short answers – Quick Revision

- **Which strengthening method does not reduce toughness?** Grain refinement.
- **Why does age hardening peak and then fall?** Shearing → Orowan transition ; over-ageing coarsens particles ($\lambda \uparrow$).
- **Why is carbon more effective than substitutional atoms?** Asymmetric tetragonal distortion ; interacts with both edge and screw dislocations.
- **What causes Lüders bands?** Locally unpinned dislocations multiply in a band ; the band front advances at the lower yield stress.
- **What does a serrated flow curve indicate?** Dynamic strain ageing (PLC effect).

13. Quick Formula Sheet – Unit II

Item	Formula
Cold work	$\%CW = (A_0 - A_f)/A_0 \times 100$; $\Delta\tau = \alpha G b \sqrt{\rho}$
Hall–Petch	$\sigma_y = \sigma_0 + k_y\, d^{-1/2}$
Solid solution	$\Delta\tau \propto c^{1/2}$ (or $c^{2/3}$)
Martensite	$c/a = 1 + 0.045\ \%C$
Orowan	$\Delta\tau = Gb/\lambda$; $\lambda \approx 2r(\sqrt{(\pi/4f)} - 1)$
Composite	$E_c = E_f V_f + E_m V_m$; $l_c = \sigma_f d/2\tau_i$
Strain ageing	$f \propto t^{2/3}$ (Harper)

14. Important Questions

- Explain strain (work) hardening and the effect of cold work on properties ; discuss recovery, recrystallisation and grain growth.
- Derive/Explain the Hall–Petch relationship. Why does grain refinement improve toughness?
- Explain solid-solution strengthening and Cottrell atmosphere. Compare interstitial and substitutional solutes.
- Explain the strengthening of steel by martensite. How does tempering modify it?
- Describe precipitation hardening of Al–Cu with the heat treatment, precipitation sequence and mechanisms (shearing vs Orowan).
- Differentiate dispersion strengthening and age hardening; explain fibre strengthening with the rule of mixtures.

- Explain the yield-point phenomenon, strain ageing and dynamic strain ageing (PLC effect).